

Design Goal:

The goal of this project is to design a portable, parent-operated restroom door locking device that can be quickly deployed (under 20 seconds) in public bathroom stalls to ensure a child's privacy when the child cannot reach or operate standard stall locks. The device must be small, lightweight, TSA-compliant, reusable (8 uses per day over a 2-day period), and effective across a wide range of public restroom door configurations. For this project, portable is defined as under 0.5 lb and $<100\text{cm}^3$, and quick deployment is defined as installation and removal both being within 20 seconds. The lock must be capable of keeping a bathroom stall door closed for at least 15 minutes, the maximum time required to accommodate a complete restroom use without interruption. Our priorities are in order of highest priority at the top and diminishing priority as it goes down.

Priorities:

1. Lock Effectiveness (High Priority)

- The lock must be able to keep the bathroom stall door closed, (within 2 degrees of perpendicular to the wall for a minimum of 15 consecutive minutes
- Effectiveness is defined as the door remaining fully closed without slipping, disengaging, or opening during the time interval.
- The 15-minute requirement is based on medical guidance for healthy bowel movements, with an added buffer of 10 minutes to account for children's slower or interrupted restroom use¹.
- This requirement supports the primary stakeholder's need for continuous privacy; failure at any point during use would compromise the child's comfort and dignity.

2. Door Variety (High Priority)

- The lock must function effectively across a wide variety of common public restroom stall doors with respect to door material, thickness, and opening orientation.
- To meet this requirement, the lock must: work on doors with thicknesses ranging from 1-2 inches²; be compatible with common stall materials, including wood, plexiglass, metal, fiberglass-enforced-plastic³; and function on doors that open both inward and outward.
- This criterion reflects the stakeholder's uncertainty regarding restroom conditions during travel and ensures that the lock can be reliably used without prior knowledge of the door type.

3. Efficacy in Public Spaces (High Priority)

- The design must be safe, legal, and practical for use in public environments, such as airport or gas-station restrooms.
- Specifically, the device must comply with TSA carry-on guidelines regarding materials, size, and shape⁴.
- The device must cause no permanent or cosmetic damage to the restroom stall doors or surrounding structures.
- This requirement is high priority because non-compliance would prevent the device from being transported or used during travel, rendering it completely ineffective for its intended purpose for use in public spaces.

4. Durability (High Priority)

- The lock must remain functional over repeated use without deformation of the device.
- The device must function effectively for a minimum of 8 uses per day over 2 consecutive days⁵ without deformation.
- Deformation is defined as a >10% alteration in length, width, height, or mass or a loss of functionality.
- This criterion aligns with the primary stakeholder's need for a reliable, reusable solution that does not require replacement after minimal use. If the lock was deformed or deemed ineffective after one use, then the stakeholder's or their children's privacy would not be protected for all public restroom trips on the vacation.

5. Weight and Volume (High Priority)

- The device should be easy to transport throughout the day, with a maximum weight of 0.5 lb and a maximum volume of 100 cm³ (based upon the proportions of an iPhone, which most can hold in their pocket⁶).
- This criterion aligns with the stakeholder's goals because they will be carrying the device around with them, so for ease of transport, it must be light and take up a small amount of space. This requirement eases the burden on the stakeholders who will have to carry the device.

6. Speed of Deployment (Low Priority)

- The design must be able to be fully installed in under 20 seconds and removed in under 20 seconds.
- While lower priority, rapid deployment improves usability and ensures the lock can be applied before the child begins using the restroom, supporting privacy and reducing

caregiver stress. With a short application period, the users will not be burdened with excess effort or annoyance.

- We based the speed of deployment criterion on our stakeholders' preferences, thus it is evidently in alignment with stakeholder goals.

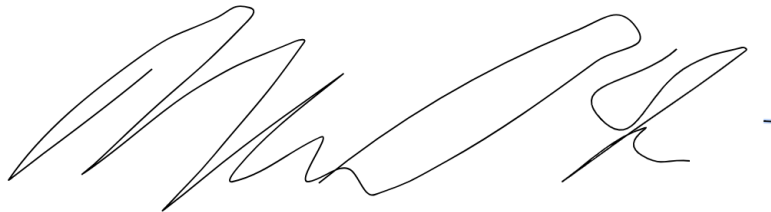
7. Low Price (Low Priority)

- The device should be affordable to create and buy.
- While no strict price ceiling is defined at this stage, the design should minimize unnecessary complexity or materials that would significantly increase cost.
- On average, the cost of a similar device is $\$12.10^7$ when choosing $n=4$ samples.
- This requirement supports broader accessibility and adoption, which supports stakeholder concerns because the stakeholders need to be able to afford the device.

Stakeholder Validation:

Stakeholder Validation of Criteria

Signature:

A handwritten signature in black ink, consisting of several loops and a long horizontal stroke, followed by a small dash.

Citations:

1. (Healthy time for a bowel movement) Jacoby, 9-26-2025, "How Long Should a Healthy Poop Take? GI Doctor Reveals the Answer", TODAY, <https://www.today.com/health/diet-fitness/how-long-normal-poop-takes-gi-doctor-tip-rcna233219>
2. (Common door thickness) MI, 10-10-2024, "Standard Door Sizes", MI Windows and Doors, <https://miwindows.com/blog/standard-door-sizes>
3. (Common door materials) Josh Smith, No Publication Date, "What are the different types of materials used for commercial bathroom partitions? Here is a Comparison", Special-Lite,

<https://special-lite.com/what-are-the-different-types-of-materials-used-for-commercial-bathroom-partitions-here-is-a-comparison/>

4. (TSA guidelines for lock materials and shape) U.S. Department of Homeland Security Transportation Security Administration, No Publication Date, USDHS, <https://www.tsa.gov/travel/security-screening/whatcanibring/all-list>
5. (Bathroom Time usage) Korin Miller, 8-29-2024, "Doctors Say This Pooping Frequency Is the Most Healthy", Prevention, <https://www.prevention.com/health/a61975059/how-often-should-you-poop/>
6. (Iphone Dimensions) Apple, No Publication Date, "iPhone 16 and iPhone 16 Plus", Apple, [https://www.apple.com/iphone-16/specs/#:~:text=Width:.7.03%20ounces%20\(199%20grams\)](https://www.apple.com/iphone-16/specs/#:~:text=Width:.7.03%20ounces%20(199%20grams))
7. (Average cost of a portable lock choosing n=4)
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 - b. (\$18.42) Home Depot, No Publication Date, "First Watch Security Chrome Travel Lock 1846", Home Depot, <https://www.homedepot.com/p/First-Watch-Security-Chrome-Travel-Lock-1846/202799703>
 - c. (\$7.86) Walmart, No Publication Date, "Protégé Portable Door Lock with Steel Plate and Travel Pouch in Black", Walmart, https://www.walmart.com/ip/Portable-Door-Lock/15893910945?wmlspartner=wlpa&selectedSellerId=0&wl13=272&qclsrc=aw.ds&adid=222222227715893910945_117755028669_12420145346&wl0=&wl1=g&wl2=c&wl3=501107745824&wl4=pla-394283752452&wl5=9013085&wl6=&wl7=&wl8=&wl9=pla&wl10=8175035&wl11=local&wl12=15893910945&veh=sem_LIA&qclsrc=aw.ds&qad_source=1&qad_campaignid=12420145346&qbraid=0AAAAADmfBlpUXkrVxZqw9UeJGc7cSeh03&qclid=CjwKCAiAmp3LBhAkEiwAJM2JUOEIn-XnW7YxTeMMz2WAE7-V3kj6ncjV0b1FUIPZOE4fVD34ssl8lXoCoDgQAvD_BwE

- d. (\$12.68) Walmart, No Publication Date, "Okeba Portable Door Lock, Travel Lock, Door Locker from Inside Extra Security Measure for Airbnb, Hotel, Home, Apartment Safety", Walmart,

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